

Completely Independent Spanning Trees on 4-Regular Chordal Rings

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SUMMARY Given a graph G , a set of spanning trees of G are completely independent spanning trees (CISTs for short) if for any vertices x and y , the paths connecting them on these trees have neither vertex nor edge in common, except x and y . Hasunuma (2001, 2002) first introduced the concept of CISTs and conjectured that there are k CISTs in any $2k$ -connected graph. Later on, this conjecture was unfortunately disproved by Péterfalvi (2012). In this note, we show that Hasunuma's conjecture holds for graphs restricted in the class of 4-regular chordal rings $CR(n, d)$, where both n and d are even integers.

key words: completely independent spanning trees, chordal rings, distributed loop networks

1. Introduction

Let G be a simple undirected graph with vertex set $V(G)$ and edge set $E(G)$. A graph G is k -connected if $|V(G)| > k$ and $G - F$ is connected for every subset $F \subset V(G)$ with $|F| < k$, where $G - F$ denotes the subgraph of G that removes all vertices of F . A tree is an acyclic and connected graph. A tree T in a graph G is a *spanning tree* if $V(T) = V(G)$. A set of k (≥ 2) spanning trees of G are said to be *completely independent spanning trees* (CISTs for short) if for any two vertices $x, y \in V(G)$, the paths joining x and y on the k trees have neither vertex nor edge in common, except x and y .

The study of CISTs was due to the work of Hasunuma in the early stage [7], [8]. He showed that there are k CISTs in the underlying graph of any k -connected line digraph [7], there are two CISTs in any 4-connected maximal planar graph [8], and there are two CISTs in the Cartesian product of any 2-connected graphs [10]. In addition, Hasunuma [8] showed that determining whether a graph admits k CISTs is NP-complete, even for $k = 2$. Moreover, he conjectured that there are k CISTs in any $2k$ -connected graph. Sadly, Péterfalvi [16] disproved this conjecture by showing that, for any $k \geq 2$, there exists a k -connected graph which

does not possess two CISTs. For more counterexamples of graphs related to interconnection networks that fail to Hasunuma's conjecture, the reader can refer to [15]. By contrast, Pai et al. [14] furnished with construction schemes for showing that there are k CISTs in complete graphs, complete bipartite graphs, and complete tripartite graphs with connectivity at least $2k$. Also, there are two CISTs in chordal rings under some restricted conditions [15] (see Lemma 1) and hypercube-variant networks [13].

Recently, the degree-based conditions that are sufficient for graphs admitting CISTs was investigated. Araki [1] first showed that a graph G on $n \geq 7$ vertices has two CISTs provided $\deg(v) \geq n/2$ for every $v \in V(G)$, where $\deg(v)$ stands for the degree of v (i.e., the number of edges incident to v). The above condition is customarily called the *Dirac's condition*, which is due to the well-know sufficient condition for Hamiltonian graphs. Afterwards, Fan et al. [5] generalized the result by showing that a graph G on $n > 8$ vertices has two CISTs if it fulfills *Ore's condition*, i.e., $\deg(u) + \deg(v) \geq n$ for every pair of nonadjacent vertices $u, v \in V(G)$. Also, Chang et al. [4] proved that G admits $\lfloor n/3 \rfloor$ CISTs provided the minimum degree $\delta(G) \geq n - 2$. Until very recently, more results related to conditions of the minimum degree are known as follows. Hong and Liu [11] showed that G admits k CISTs if $\delta(G) \geq (k - 1/k)n$, where $n \geq 4k - 1$. Hasunuma [9] showed that G admits $\lfloor n/k \rfloor$ CISTs if $\delta(G) \geq n - k$, where $3 \leq k \leq n/2$ and $n \geq 7$. Note that both results generalize the result of [1], however, the latter is stronger than the former. To see this, by setting $k = 3$ in the latter, it implies that G admits $\lfloor n/3 \rfloor$ CISTs if $\delta(G) \geq n - 3$ for $n \geq 7$, but the former fails the assertion because the setting $k = \lfloor n/3 \rfloor$ conflicts $n \geq 4k - 1$ for $n \geq 12$.

In this note, we try to keep on the exploration of CISTs in chordal rings (which will be defined later in Sect. 2). A synonym of chordal rings is called the *distributed loop networks*, which were introduced in [2], [3] to enhance the reliability and fault-tolerance of ring networks. Thus, this family of graphs becomes the one of the most popular topologies of interconnection networks. We refer to [6], [12], [17] for more results on independent spanning trees (a weakness of the notion of CISTs) of chordal rings.

The remaining part of this note is organized as follows. Section 2 gives the definition of chordal rings and existing results of CISTs. Section 3 presents our constructing schemes of CISTs on chordal rings under certain conditions. Finally, a concluding remark is given in the last section.

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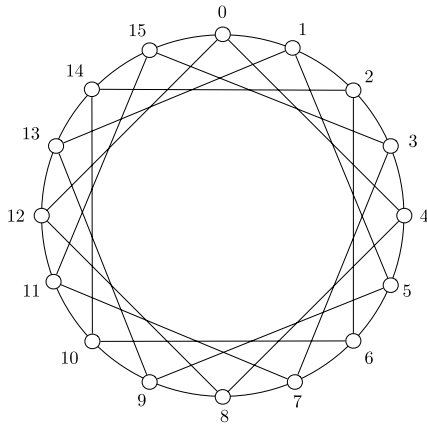


Fig. 1 Chordal ring $CR(16, 4)$.

2. Preliminaries

A *chordal ring*, denoted by $CR(n, d)$, is a graph with vertex set $V = \{0, 1, 2, \dots, n-1\}$ and edge set $E = \{(u, v) \mid u, v \in V \text{ and } (u-v) \equiv 1 \text{ or } d \pmod{n}\}$. For notational convenience, all arithmetics applied to a chordal ring are taken modulo n . To ensure that every vertex has degree 4 (i.e., a 4-regular chordal ring), we assume $n \geq 5$ and $d < n/2$. Hence, we say that the four edges $(u, u-1)$, $(u, u+1)$, $(u, u-d)$, $(u, u+d)$ incident to a vertex $u \in V$ are *hops* $-1, +1, -d$ and $+d$, respectively. For instance, a chordal ring $CR(16, 4)$ is shown in Fig. 1.

Pai et al. [15] obtained the following result of CISTs for chordal rings.

Lemma 1: (See [15].) Let $k \geq 4$ be an even integer and $j \in \{0, 1, 2, 3, 4\}$. If $n = k(d-1) + j$, there are two CISTs in $CR(n, d)$.

We will find it convenient to adopt the following notation. For a graph G and $U \subseteq V(G)$, the subgraph induced by U is denoted by $G[U]$. For disjoint subsets $U_1, U_2 \subseteq V$, the graph $B(U_1, U_2)$ is defined to be a bipartite graph with bipartition U_1 and U_2 , and edge set $\{(x, y) \in E(G) : x \in U_1, y \in U_2\}$.

Araki [1] gave the following characterization for a graph admitting k CISTs.

Theorem 2: (See [1].) A graph G admits k CISTs if and only if there is a partition of $V(G)$ into $V_1, V_2, \dots, V_k$ such that $G[V_i]$ is connected for $i \in \{1, 2, \dots, k\}$ and $B(V_i, V_j)$ has no tree component for distinct $i, j \in \{1, 2, \dots, k\}$.

Note that the partition of $V(G)$ described in Theorem 2 is called a *CIST-partition* of G .

3. Main Results

In what follows, we let $G = CR(n, d)$ with vertex set V .

Lemma 3: There exist two CISTs in $CR(n, d)$ for $n \geq 5$ and $d = \lceil n/2 \rceil - 1$.

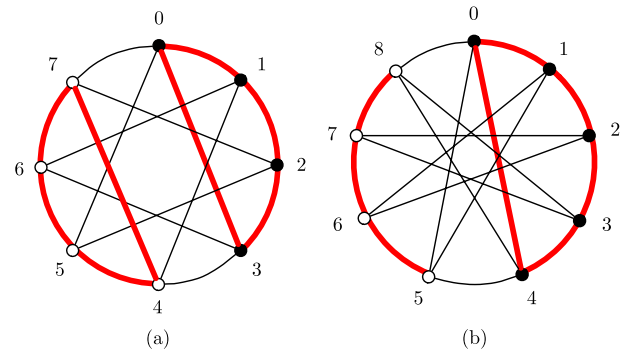


Fig. 2 Illustrations of Lemma 3: (a) $CR(8, 3)$ and (b) $CR(9, 4)$.

Proof : Since $n \geq 5$, it is clear that $d \geq 2$. Let V be partitioned into $V_1 = \{0, 1, \dots, d\}$ and $V_2 = V \setminus V_1$. That is, $\|V_1\| - \|V_2\| \leq 1$. Obviously, if n is even, both $G[V_1]$ and $G[V_2]$ form a cycle of length $d+1$ (see Fig. 2(a) as an example of $n = 8$). By contrast, if n is odd, $G[V_1]$ is a cycle of length $d+1$ and $G[V_2]$ is a path of $n-d-1$ vertices (see Fig. 2(b) as an example of $n = 9$). Thus, both $G[V_1]$ and $G[V_2]$ are connected. Since $CR(n, d)$ is 4-regular and every vertex in $G[V_1] \cup G[V_2]$ has degree at most 2, it follows that $\deg(x) \geq 2$ for every vertex x in $B(V_1, V_2)$, and thus $B(V_1, V_2)$ has no tree component. By Theorem 2, $\{V_1, V_2\}$ forms a CIST-partition of $CR(n, d)$ and the lemma follows. $\square$

As a remark, after the submission of this note, a referee pointed out that the same statement of Lemma 3 has been already mentioned in [9]. In that paper, two graphs $H(n, k)$ and $H^*(n, k)$ are defined and it is shown that $H(n, k)$ and $H^*(n, k)$ have $\lfloor n/k \rfloor$ CISTs. Note that $H(n, \lfloor n/2 \rfloor) = CR(n, \lceil n/2 \rceil - 1)$ for odd n and $H^*(n, n/2) = CR(n, \lceil n/2 \rceil - 1)$ for even n .

Lemma 4: There exist two CISTs in $CR(n, d)$ for even integers $n \geq 5$ and $d < n/2$.

Proof : Recall that $d < n/2$ for $CR(n, d)$. Since n and d are even integer, $n \geq 2d+2$. Let V be partitioned into $V_1 = \{0, 1, \dots, d-1\} \cup U$ and $V_2 = \{d, d+1, \dots, 2d-1\} \cup W$ with equal size, where $U = \{2d, 2d+2, \dots, n-4, n-2\}$ and $W = \{2d+1, 2d+3, \dots, n-3, n-1\}$. Since $n \geq 2d+2$, we have $1 \leq |U| = |W| = (n-2d)/2$. This indicates that both U and W are nonempty. Let P_1 be the subgraph of $G[V_1]$ induced by $\{0, 1, \dots, d-1\}$ and P_2 be the subgraph of $G[V_2]$ induced by $\{d, d+1, \dots, 2d-1\}$. Clearly, P_1 and P_2 are paths of length $d-1$. We now show that, for each vertex $x \in U$, there is a path joining x and a vertex of P_1 in $G[V_1]$. Let $y = (x+d) \bmod n$, i.e., x is adjacent to y via an edge with hop $+d$. Since both n and d are even integers, x and y have the same parity. There are two cases as follows:

Case $n \leq 3d$ (i.e., $|U| \leq |V(P_1)|/2 = d/2$): Since $2d \leq x \leq n-2$ and $y = (x+d) \bmod n$, we have $0 \leq 3d-n \leq y \leq d-2$. This shows that $y \in V(P_1)$, and thus $G[V_1]$ is connected. See Fig. 3(a) for an illustration of the subgraphs $G[V_1]$ and $G[V_2]$.

Case $n \geq 3d+2$: Let $U = U_1 \cup U_2$, where $U_1 = \{n-d, n-d+2, \dots, n-2\}$ and $U_2 = \{2d, 2d+2, \dots, n-d-2\}$.

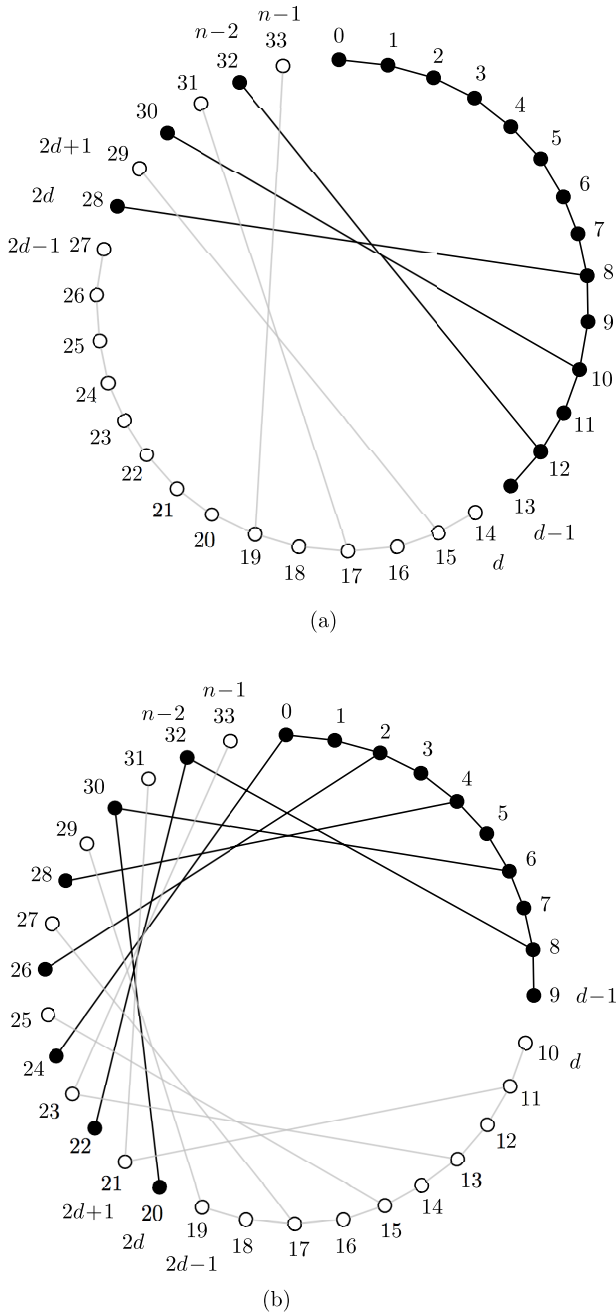


Fig. 3 Illustrations of Lemma 4: subgraphs $G[V_1] \cup G[V_2]$ in chordal rings (a) $CR(34, 14)$ and (b) $CR(34, 10)$, respectively.

If $x \in U_1$, then $0 \leq (x+d) \bmod n \leq d-2$, and thus $y \in V(P_1)$. If $x \in U_2$, then $y \equiv x+d \pmod{n} \in \{3d, 3d+2, \dots, n-2\} \subset \{2d, 2d+2, \dots, n-2\} = U$. Thus, if $y \in U_1$, then y is adjacent to a vertex of P_1 . Otherwise, y must be connected to a vertex of P_1 via a path of length at least two. Hence, $G[V_1]$ is connected. See Fig. 3(b) for an illustration of the subgraphs $G[V_1]$ and $G[V_2]$.

A similar argument shows that, for each vertex $x \in W$, there is a path joining x and a vertex of P_2 in $G[V_2]$, and hence $G[V_2]$ is connected.

In the following, we show that $B(V_1, V_2)$ has no tree

component. We first observe that $B(V_1, V_2)$ contains the path $P = (2d-1, 2d, 2d+1, \dots, n-1, 0)$ with vertices being alternated between V_1 and V_2 . Moreover, P together with $(0, d, d-1, 2d-1)$ forms a cycle in $B(V_1, V_2)$, where we denote the cycle by C . Clearly, if $d = 2$, then $|V(C)| = n$ and thus we are done. For $d \geq 4$, we claim that, for each vertex $x \in V \setminus V(C)$, there is a path joining x and a vertex of C in $B(V_1, V_2)$. Let

$$H^- = \{1, 3, 5, \dots, d-3\},$$

$$H^+ = \{2, 4, 6, \dots, d-2\},$$

$$L^- = \{d+1, d+3, d+5, \dots, 2d-3\},$$

and

$$L^+ = \{d+2, d+4, d+6, \dots, 2d-2\}.$$

Clearly, $V \setminus V(C) = H^- \cup H^+ \cup L^- \cup L^+$. Note that vertices in $H^- \cup L^-$ (respectively, $H^+ \cup L^+$) have the same parity. Consider the following two cases:

Case $n \geq 3d-2$: For $x \in H^-$, let $y = (x-d) \bmod n$ be a vertex adjacent to x . Since $n \geq 3d-2$ and $1 \leq x \leq d-3$, it implies $2d-1 \leq n+1-d \leq (x-d) \bmod n \leq n-3$. Thus, we have $y \in V(P)$. For $x \in L^+$, let $y = x+d$ be a vertex adjacent to x . Since $2d+2 \leq x+d \leq 3d-2 \leq n$, it follows that $y \in V(P)$. For $x \in H^+$, let $y = x+d$ be a vertex adjacent to x . It is clear that $y \in L^+$. For $x \in L^-$, let $y = x-d$ be a vertex adjacent to x . It is clear that $y \in H^-$. Thus, our assertion is achieved. An illustration is given in Fig. 4(a) for showing that $B(V_1, V_2)$ has no tree component.

Case $n \leq 3d-4$: For $x \in H^-$, let $y = (x-d) \bmod n$ be a vertex adjacent to x . Since $1 \leq x \leq d-3$, it implies $n+1-d \leq n+x-d = y \leq n-3$. If $y \geq 2d-1$, it is clear that $y \in V(P)$. Otherwise, since $n \geq 2d+2$, we have $d+3 \leq n+1-d \leq y < 2d-1$. Since x and y have the same parity, it follows that $y \in L^-$. For $x \in L^-$, let $y = (x-d) \bmod n$ be a vertex adjacent to x . It is clear that $y \in H^-$. This shows that every vertex of $H^- \cup L^-$ is connected to a vertex of C by a path. By symmetry, a similar argument shows that the assertion still holds for vertices in $L^+ \cup H^+$. As a result, $B(V_1, V_2)$ contains no tree component. An illustration is given in Fig. 4(b) for showing this case.

From above, it shows that $\{V_1, V_2\}$ forms a CIST-partition of $CR(n, d)$. By Theorem 2, the lemma follows. $\square$

According to Lemmas 1, 3 and 4, we summarize existing results of CISTs for chordal rings as follows.

Theorem 5: There are two CISTs in a 4-regular chordal ring $CR(n, d)$ if one of the following conditions is fulfilled:

- (i) $n = k(d-1) + j$ for $k \geq 4$ and $0 \leq j \leq 4$;
- (ii) $n \geq 5$ and $d = \lceil n/2 \rceil - 1$; and
- (iii) Both $n \geq 5$ and $d < n/2$ are even integers.

4. Concluding Remarks

In this note, we show that there exist two CISTs of $CR(n, d)$

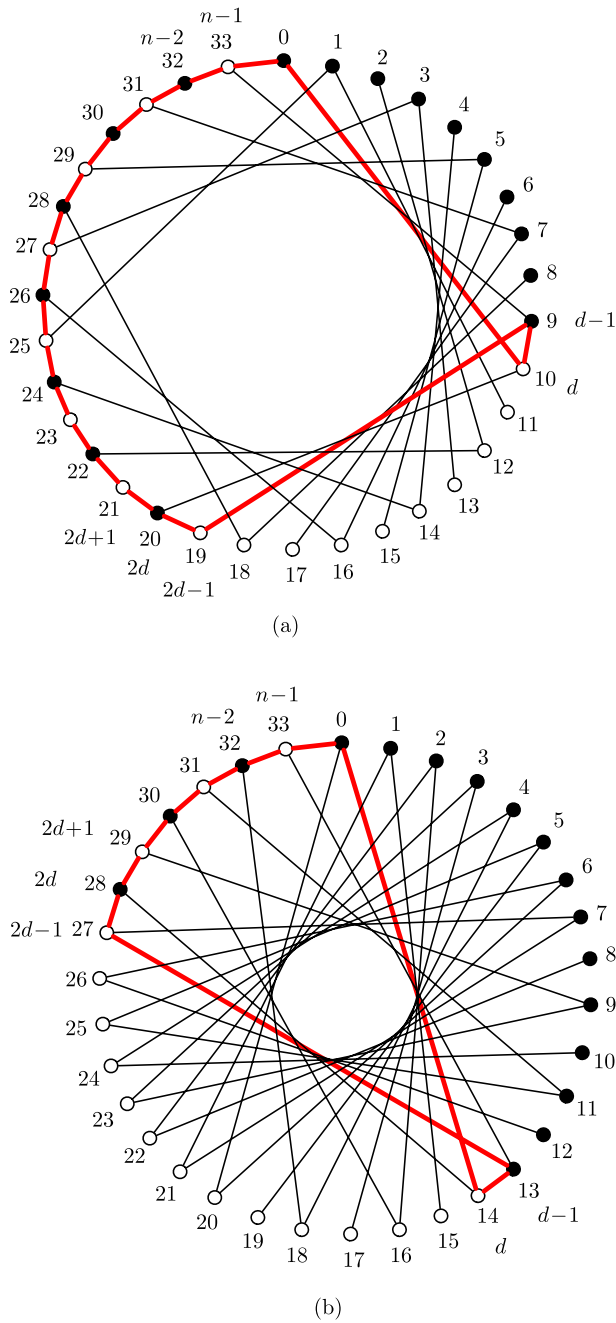


Fig. 4 Illustrations of Lemma 4: subgraphs $B(V_1, V_2)$ in chordal rings (a) $CR(34, 10)$ and (b) $CR(34, 14)$, respectively.

when both n and d are even. At present, the construction of two CISTs in 4-regular chordal rings has not been solved completely. So far very little effort has been devoted to the construction of CISTs for interconnection networks (e.g., high dimensional tori, hypercubes and their variants). It is worthy to study in this direction as a future work.

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